Lab Report 03

Course title: Computer Network Laboratory
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Submitted to-

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Title: Dynamic Host Configuration Protocol (DHCP) Using Router and Server

Objective:

- To understand the basic concept of DHCP.
- To configure a router as a DHCP server.
- To configure a server as a DHCP server.
- To automatically assign IP addresses to PCs using DHCP.
- To verify the DHCP configuration in Cisco Packet Tracer.

Description:

Part 1: DHCP Using Router

In this experiment, one 2911 router, one 2960 switch and six PCs were used. All the devices were connected using cables.

First the router was accessed through the CLI. The GigabitEthernet 0/0 interface was configured with the IP address 192.168.1.1 and subnet mask 255.255.255.0. The following commands were used:

```
en
conf t
int gig0/0
ip address 192.168.1.1 255.255.255.0
no sh
exit
```

After configuring the router interface, a DHCP pool named LAN-1 was created using the following commands:

```
ip dhcp pool LAN-1
net 192.168.1.0 255.255.255.0
default-router 192.168.1.1
```

The ip dhcp pool LAN-1 command created the DHCP pool. The network command specified the network from which the IP addresses would be assigned and the default-router command specified 192.168.1.1 as the default gateway.

Finally, the IP configuration of each PC was opened and DHCP was selected. The PCs automatically received their IP addresses from the router. This confirmed that the router was successfully configured as a DHCP server.

Part 2: DHCP Using Server

In the second part, a server was used instead of the router to provide DHCP service. One server, one switch and six PCs were used and connected together.

First, the server was configured with the IP address 192.168.2.1 and subnet mask 255.255.255.0 through Config - FastEthernet0.

Then, the DHCP service was enabled from Services - DHCP. A DHCP pool was configured with the starting IP address 192.168.2.2 and subnet mask 255.255.255.0. The DHCP configuration was then added.

After the DHCP service was configured, the IP configuration of all PCs was opened and DHCP was selected. Each PC automatically received an IP address from the server. This confirmed that the server was successfully providing DHCP services to the connected PCs.

Connection:

Part 1: DHCP Using Router

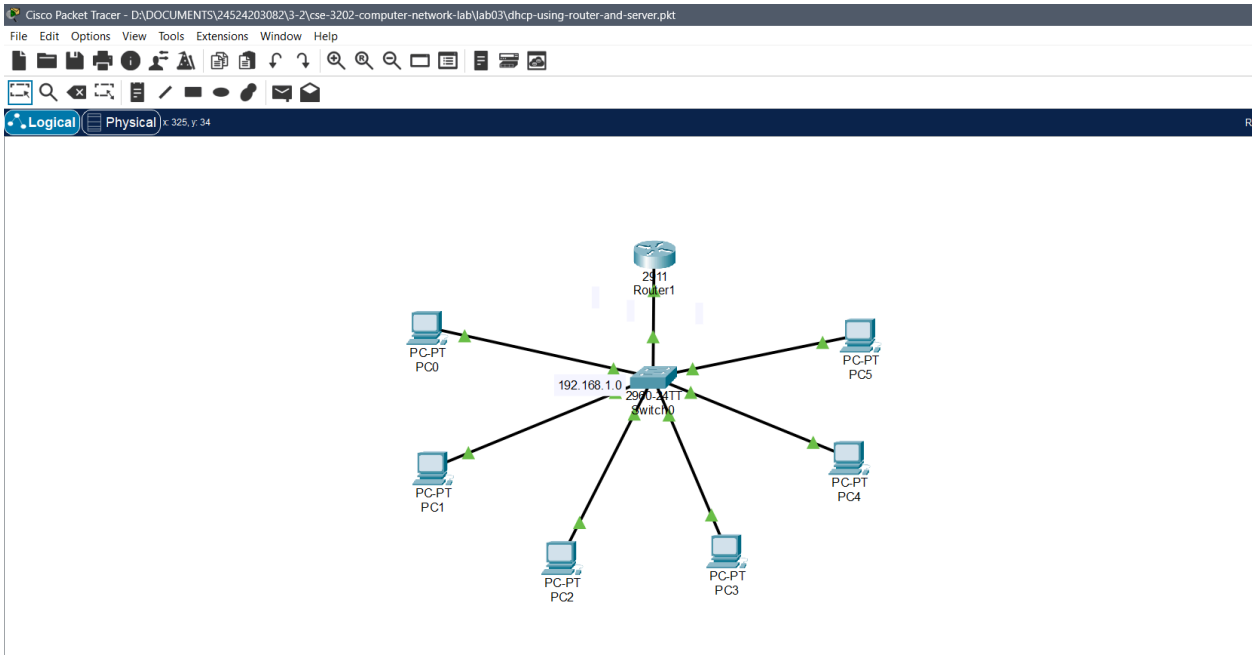


Figure 1: Network Topology for DHCP Using Router

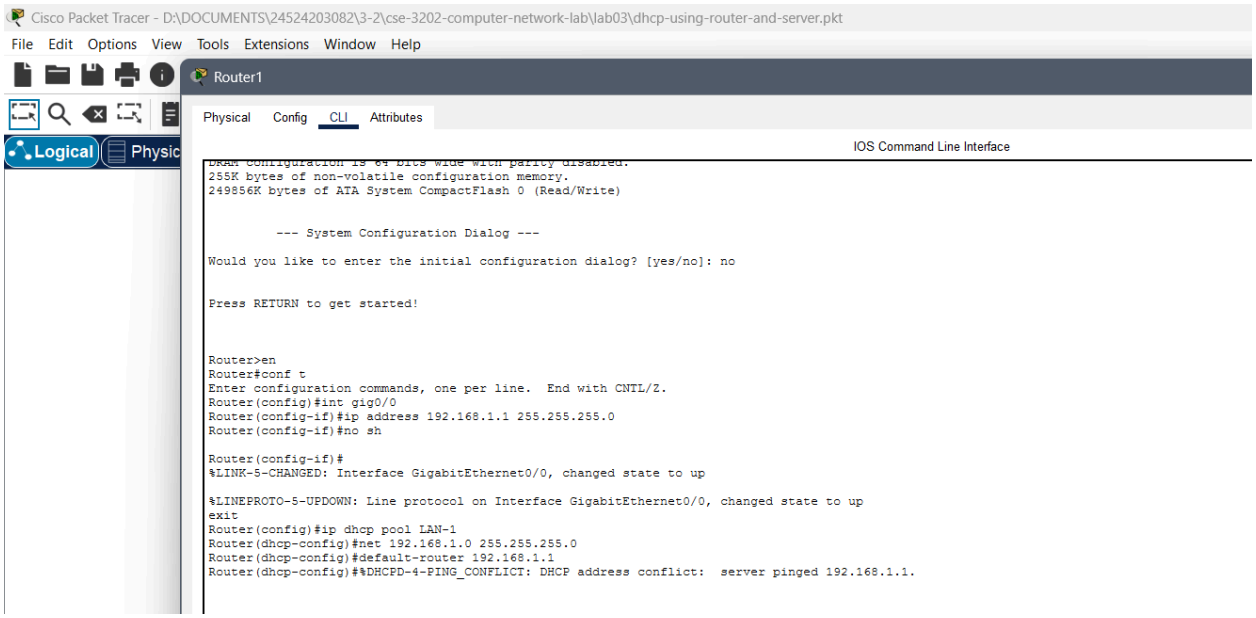


Figure 2: Router CLI Configuration

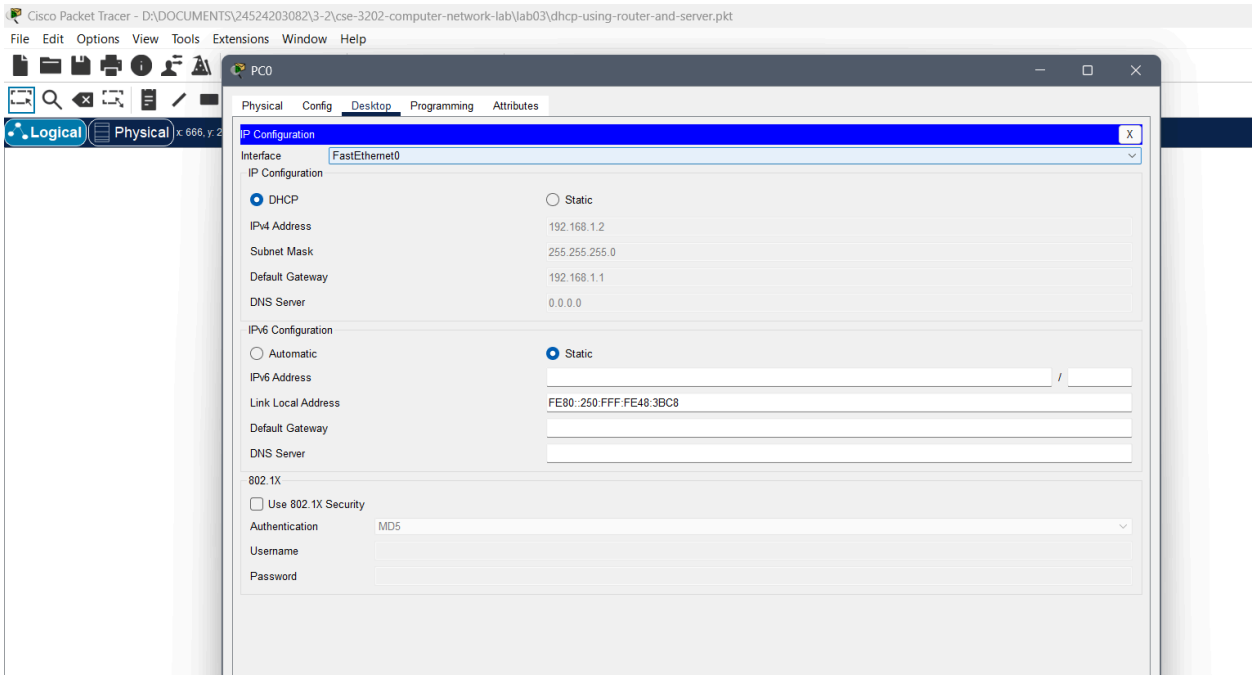


Figure 3: Automatically Assigned IP Address on PC

Part 2: DHCP using Server

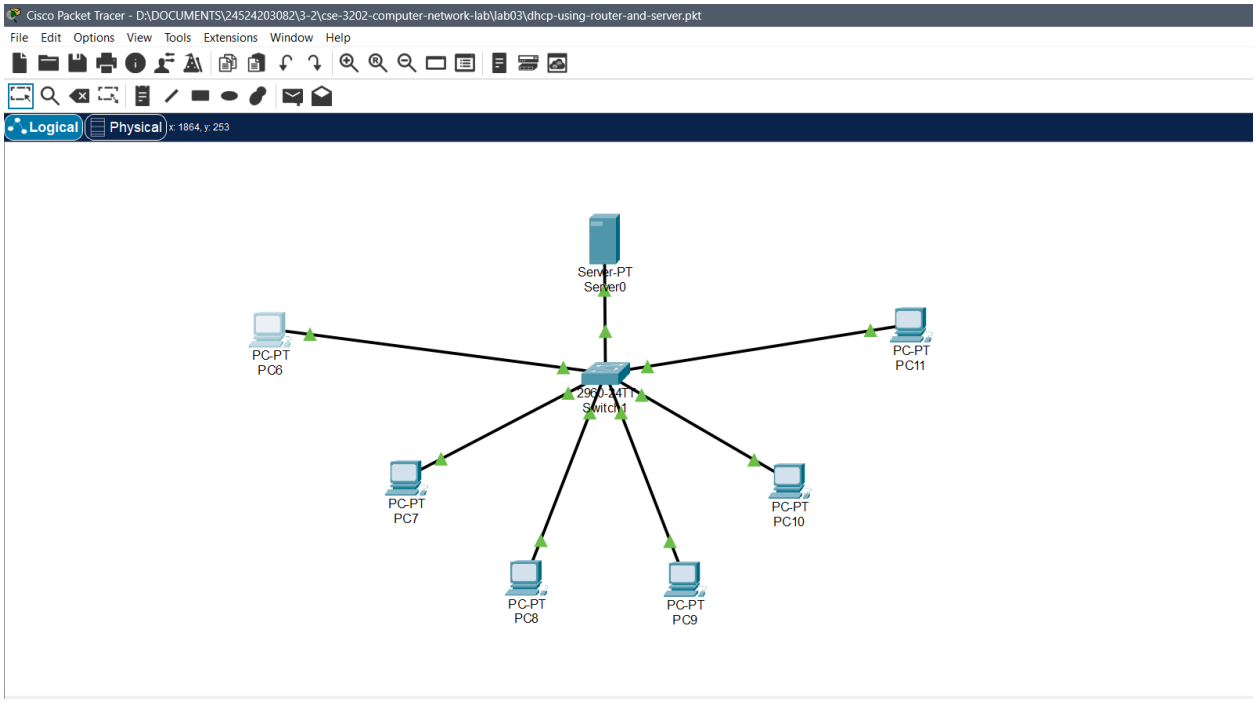


Figure 4: Network Topology for DHCP Using Server

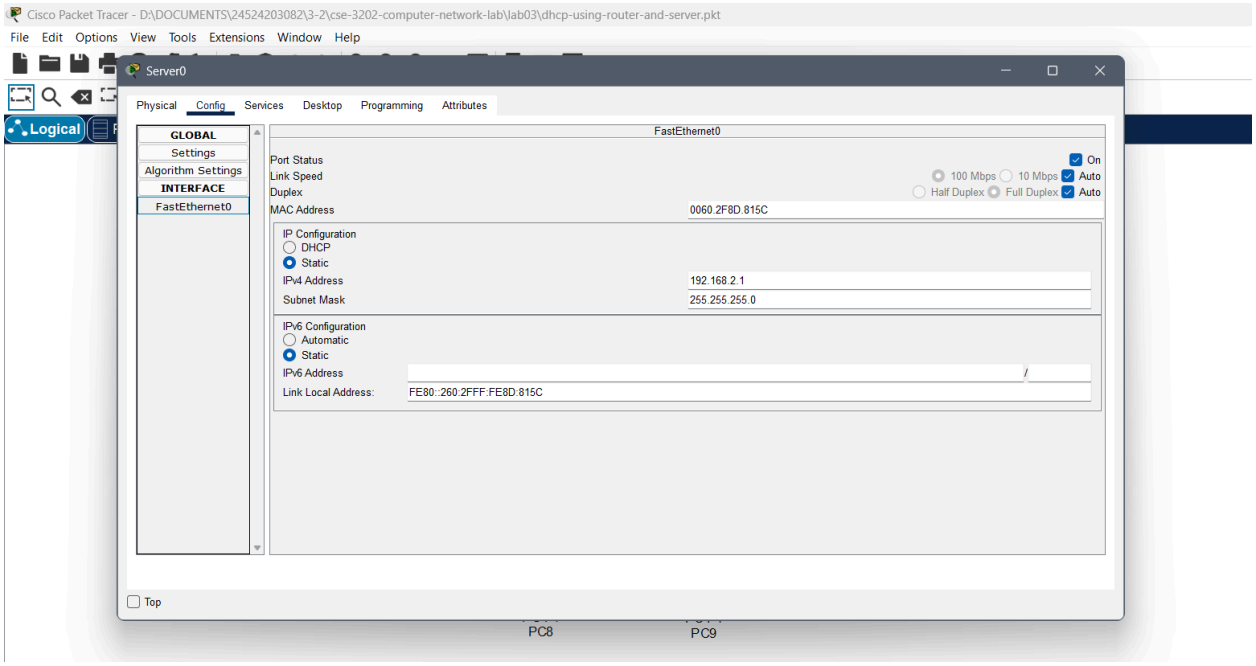


Figure 5: Server FastEthernet0 IP Configuration

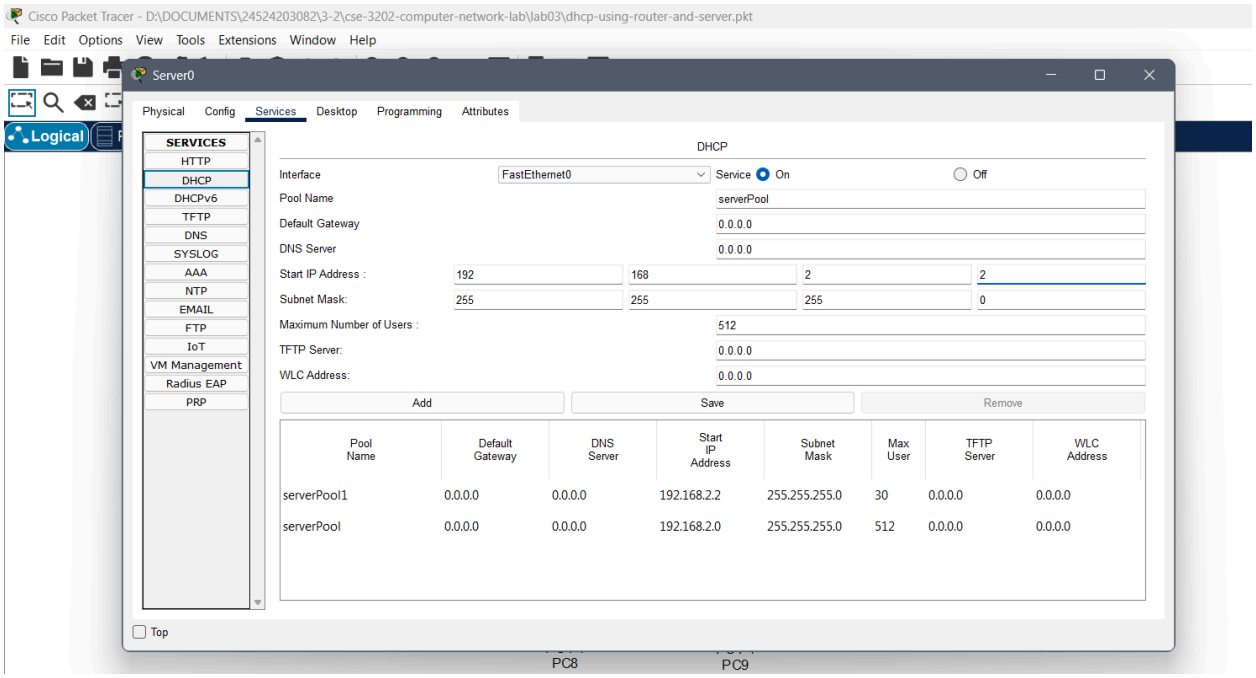


Figure 6: DHCP Service Configuration on Server

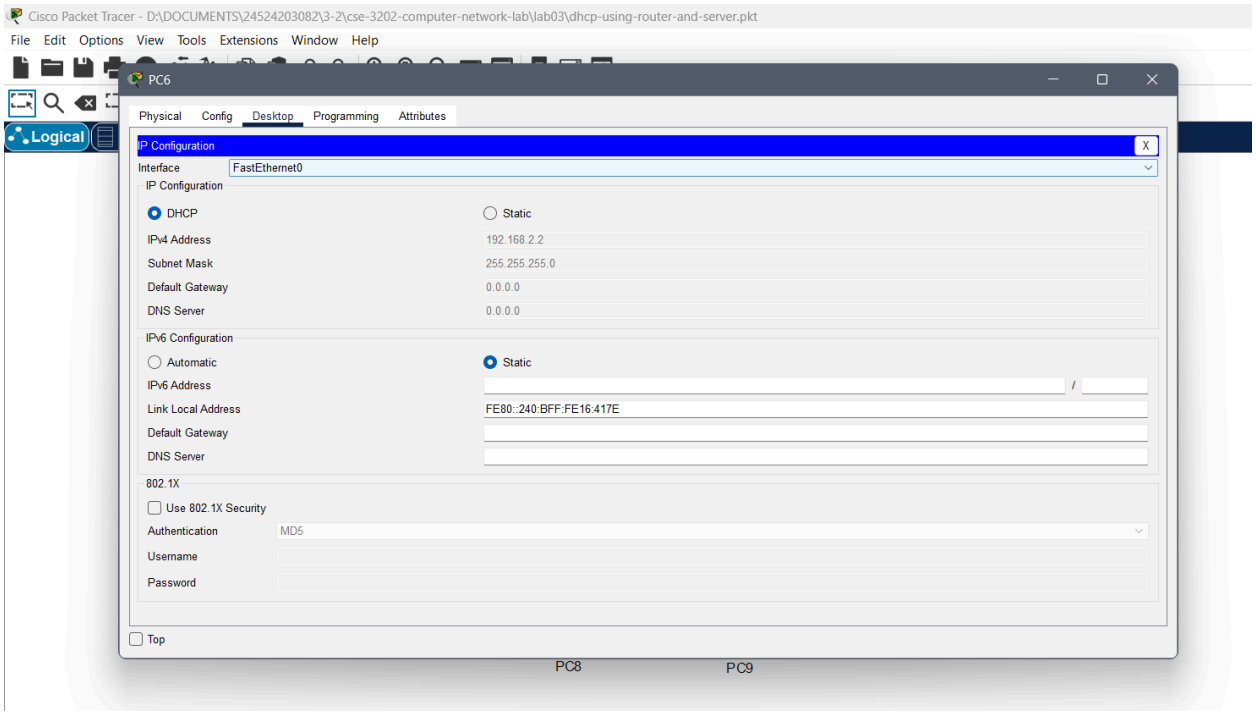


Figure 7: Automatically Assigned IP Address from DHCP Server